The Khronometric Equation of State: Jacobson's Derivation, Written Out for Event Density's Luminal-Cone Gravity

Allen Proxmire

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Preamble — what this article does and does not claim

1. **This writes out a derivation whose verdict is already established, not a new verdict.** The companion paper `Paper_GravityAsEquationOfState` §6 states, at literature-grounded tier, that ED's local Jacobson derivation is *cleanly equilibrium and reaches the full Einstein equation*. Nobody — in ED or in the Einstein-aether literature — has actually **written out** Jacobson's local-Rindler `δQ = TdS` derivation for a khronometric theory. This article does that, for ED's specific *luminal-cone* class, making the verdict explicit and checkable.
2. **The Jacobson core is derived here; the aether-specific results are cited, not re-derived.** The local-Rindler Clausius steps (Raychaudhuri focusing, the boost Killing flux) are worked in full. The three aether-specific facts the derivation rests on — (i) the Wald entropy is pure area, (ii) the non-area aether entropy is reabsorbable in a local aligned single-scale wedge, (iii) the shear channel is unmodified on the luminal family — are **taken from the literature** (cited inline) and assembled, not independently re-derived.
3. **ED's results are conditional derivations from thirteen posited primitives**, never derivations from nothing, and the framework is not experimentally confirmed.
4. **The "fourth face" is NOT claimed.** That the khronon's kinematic footprint (α_1) and its dissipative footprint (a bulk entropy-production term) are the *same* function of the one coupling is an **open conjecture**, not a result. What is banked is weaker and stated as such (§6): both are `O(λ_local)`, neither leaks at `O(1)`, and the exact-same-power question needs a closed-form aether entropy that even the 2026 literature has not rendered.
5. **Tiers are marked per claim.** Determination (C) (clean equilibrium $\to$ full Einstein) is *literature-grounded*; the `O(λ_local)` powers rest on a flux-prefactor *inference*; the absolute seventy-orders magnitude is *inherited* from GR-IV's estimate-tier `λ_local`.

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1. What Event Density is, and the class this runs in

Event Density is a discrete, relational substrate: a participation graph whose channels carry a bandwidth `b ≥ 0` and a phase, with one process primitive — **commitment** (P11), the irreversible act that writes the arrow of time into the law. Gravity is emergent (the metric is the reading of the bandwidth field, `g ~ 1/b`, `Paper_GR-I`), and ED's gravitational class is **khronometric**: Einstein's two tensor modes plus one propagating scalar, the *khronon* $u_\mu = \partial_\mu T/|\partial T|$, the arrow made geometric (`Paper_GR-II`).

The emergent horizon is the $b \to 0$ locus, an A1 capacity-zero causal cut (Paper_GR-III §7.4).

The one feature that makes this derivation tractable is that **both of ED's gravitational cones are luminal**: $c_T = c$ (the single P05 transport cone) and $c_s = c$ at leading order (the khronon; the commitment reserve is dissipative, not a second kinetic cone, Paper_GR-III §6). In the standard three-coupling khronometric parametrization (α, β, λ) — equivalently the Einstein-aether (c_1, c_2, c_3, c_4) with hypersurface-orthogonality removing the spin-1 mode — luminality is two independent constraints ($c_T = c \implies \beta = 0$; $c_s = c \implies \alpha = \lambda/(1+2\lambda)$), collapsing the coupling space to **one dimension**, coordinatized by $\lambda \equiv \lambda_{local}$ (Paper_GR-IV §3; verified analytically against Ramos–Barausse arXiv:1811.07786, Blas–Lim arXiv:1412.4828, Foster–Jacobson gr-qc/0509083). On that family, $\alpha_2 = 0$ exactly and $\alpha_1 = -4\lambda_{local}$. This 1D collapse is what lets a single coupling govern every khronometric footprint below.

This article is the technical companion to Paper_GravityAsEquationOfState: that paper grounds Jacobson's *inputs* and asserts the equilibrium verdict; this one writes out the *derivation* that produces it.

2. Jacobson's derivation, and the four places Lorentz enters

Jacobson (1995) derives the Einstein equation as an equation of state. Through any point p, boost into a local inertial frame and consider a small patch of a **local Rindler horizon** — a null congruence with generators k^a, affine parameter λ, approximate boost Killing field $\chi^a = -\kappa\lambda\, k^a$. Three inputs:

- **Entropy** proportional to horizon area, $S = \eta A$, so $dS = \eta\, \delta A$;
- **Unruh temperature** $T = \kappa/2\pi$ (the local Rindler temperature of the boost vacuum);
- **Heat** = boost energy flux across the horizon, $\delta Q = \int T_{ab}\chi^a \, d\Sigma^b = -\kappa \int \lambda \, T_{ab} k^a k^b \, d\lambda \, dA$.

The area change follows from **Raychaudhuri focusing**, $d\theta/d\lambda = -\tfrac12 \theta^2 - \sigma^2 - R_{ab}k^a k^b$. Choosing p so that $\theta = \sigma = 0$ there, to first order $\theta = -\lambda R_{ab}k^a k^b$, hence

$$\delta A = \int \theta \, d\lambda \, dA = -\int \lambda \, R_{ab} k^a k^b \, d\lambda \, dA.$$

Imposing Clausius $\delta Q = T\, dS$ for **all** local Rindler horizons (all null k^a) forces $T_{ab}k^a k^b = (\eta/2\pi)\, R_{ab}k^a k^b$ for every null k, hence $T_{ab} = (\eta/2\pi)(R_{ab} + f g_{ab})$; conservation and the Bianchi identity fix $f = -R/2 + \Lambda$, giving the **Einstein equation** with $2\pi/\eta = 8\pi G$.

The four Lorentz-dependent steps (the audit the khronometric case must survive): (1) the Unruh temperature assumes the vacuum is *thermal with respect to boosts*; (2) the boost Killing field assumes an *approximately flat region with Poincaré symmetry*; (3) "all null directions" assumes an *isotropic light cone / no preferred frame*; (4) the entanglement origin of S assumes *Lorentz-invariant vacuum correlations*. A preferred foliation threatens each. The rest of this article shows each survives on ED's luminal family, and identifies exactly where the khronon does leave a mark.

3. The luminal cones keep the kinematic scaffolding (steps 1–3)

The kinematic worries are benign, and the reason is uniform: **on ED's luminal family the causal structure is the standard light cone.**

- **No universal-horizon complication.** The notorious "universal horizon" of khronometric/Hořava gravity — a khronon-trapping surface distinct from the metric horizon — exists only because the khronon can propagate *superluminally*, so the metric light-cone is not the true causal boundary (Barausse–Jacobson–Sotiriou arXiv:1104.2889; Blas–Sibiryakov arXiv:1110.2195). ED's khronon is *luminal* (`c_s = c`), so the metric horizon **is** the causal horizon and no separate universal horizon arises. Step (3)'s "all null directions" is the ordinary null cone.

- **The Unruh temperature survives (step 1).** What breaks horizon thermality under Lorentz violation is dispersion *nonlinearity*, not the preferred frame as such: linear/luminal dispersion preserves the KMS/thermal structure (arXiv:2102.08944), while nonlinear or superluminal dispersion breaks it (Campo–Obadia arXiv:1003.0112). ED's dispersion is linear and luminal, so `T = κ/2π` holds. (Within ED this temperature is itself read off the near-horizon Rindler geometry via the Euclidean-continuation route, with the standard honesty pin that a commitment-statistics, continuation-free derivation is open — `BH_Thermal2Pi` §4b.)
- **$\alpha_2 = 0$ exactly (step 2).** The second preferred-frame parameter vanishes identically on the luminal family (`Paper_GR-IV` §3, literature-verified), so the boost structure step 2 relies on is undeformed at that order. The only surviving kinematic footprint is `α₁ = −4λ_local`, a velocity-dependent correction suppressed ∼70 orders by sparse becoming (`Paper_GR-IV` §6).

So steps 1–3 pass, and the "khronometric-vs-Lorentz obstruction" is **real but benign at the kinematic level**. The genuine question is step (4): the entropy.

4. The horizon entropy is pure area (the equilibrium enabler, step 4)

Eling, Guedens and Jacobson (gr-qc/0602001) sharpened the equation-of-state program: `δQ = TdS` holds in the tidy **equilibrium** form if and only if the horizon entropy is **pure area**; any non-area piece forces a non-equilibrium Clausius relation `dS = δQ/T + d_iS` with an internal-production term `d_iS ≥ 0`. So everything turns on whether the khronon adds a non-area entropy to ED's horizon.

It does not, locally. Two facts, both from the literature, settle it:

- **The Wald entropy is pure area.** The Wald entropy density is built from $\partial L/\partial R_{abcd}$. Only the Einstein–Hilbert term carries the Riemann tensor; the khronon/aether kinetic terms `(\nabla u)^2` contain no explicit `R_{abcd}`, so they contribute **zero** to the Wald density. The horizon entropy is `S = A/(4G_æ)` with `G_æ = G_N/(1 − c_{14}/2)` — the couplings only **renormalize the area coefficient**, adding no non-area piece (Brustein–Gorbonos–Hadad arXiv:0712.3206; arXiv:2606.27437). (The naive first law is singular at the bifurcation surface, where the aether unit vector is ill-defined; the covariant-phase-space treatment makes it rigorous and confirms the area piece — Foster gr-qc/0510125; arXiv:2603.28851.)
- **The non-area "aether entropy" is reabsorbable in the local wedge.** There *is* a candidate non-area aether term, but arXiv:2603.28851 §6.3 shows that in a **single-scale, asymptotically-flat** geometry it is reabsorbed into the area coefficient "by a simple rescaling," becoming genuinely independent "only in the presence of additional scales (such as a cosmological constant)"; arXiv:2606.27437 shows it appears only when the aether is **misaligned** with the horizon-generating Killing vector. Jacobson's construction uses an **aligned, asymptotically-flat, single-scale** (no Λ) local Rindler wedge — precisely the case where the aether entropy reabsorbs and does not enter.

Determination (C). In ED's local Rindler wedge the horizon entropy is pure area, so the Clausius relation is the tidy **equilibrium** one, and Jacobson's derivation runs unchanged (with `η = 1/4G_æ`). *(Literature-grounded — the Wald pure-area result and the reabsorption are drawn from the cited 2026 covariant-phase-space literature, assembled via the R1–R6 analysis, not independently re-derived here; see §7 tiers.)*

5. The explicit equation of state: clean equilibrium $\to$ full Einstein

Assemble §§2–4. With pure-area entropy `dS = η\, δA`, `η = 1/(4G_æ)`, temperature `T = κ/2π`, and heat flux from the **total** stress (matter plus the khronon's own stress, `T_{ab}^{\rm tot} = T_{ab}^{\rm m} + T_{ab}^{æ}`), Clausius `δQ = T dS` reads

$$-\kappa \int \lambda\, T^{\rm tot}_{ab} k^a k^b \, d\lambda \, dA \;=\; \frac{\kappa}{2\pi}\, \eta \left(- \int \lambda \, R_{ab} k^a k^b \, d\lambda \, dA \right).$$

Demanding it for all null `k^a` gives `T_{ab}^{\rm tot} k^a k^b = (\eta/2\pi) R_{ab} k^a k^b` for every null `k`, hence, with conservation and Bianchi as in §2,

$$\boxed{R_{ab} - \tfrac{1}{2} R \, g_{ab} + \Lambda g_{ab} \;=\; 8\pi G_{\rm æ} \, (T^{\rm m}_{ab} + T^{\rm æ}_{ab}).}$$

This is the Einstein-aether field equation. Now use Determination (C): because the horizon entropy is pure area (§4), the aether adds **no non-area entropy and no non-equilibrium production term** to the local Clausius relation. It enters only as an ordinary source stress `T_{ab}^{æ}` on the right-hand side and through the renormalized coefficient `G_æ`. In Jacobson's sense (geometry = total stress), the local wedge relation is therefore of **Einstein form with the aether counted among the sources** — the clean equilibrium equation of state, not a modified or non-equilibrium one. It is **not** vacuum General Relativity: `T_{ab}^{æ} ≠ 0` and the khronon carries its own field equation; what is "full Einstein" is the *form and equilibrium class* of the relation, not the disappearance of the aether from the right-hand side. The genuinely khronometric, non-Einstein content (the non-area entropy, the bulk dissipation of §6) appears only when a second scale (`Λ`) or misalignment is present — i.e. at the *global/cosmological* level — or at `O(λ_local)` in the dissipative channel. Locally:

> **ED reaches the full Einstein equation via Jacobson's clean equilibrium derivation.** This is the same class as General Relativity itself: GR's own local equation of state is equilibrium up to the universal shear-viscosity term (the `σ^2` Jacobson drops = Hartle–Hawking tidal heating; Chirco–Liberati PRD 81, 024016), and ED carries exactly that same term and no more at the tensor level (§6). ED is *not* below GR in equilibrium quality, and it is *not* above it.

This is the second independent route to Einstein in ED — the thermodynamic one — meeting the dynamical bandwidth-rule route (`ḃ = D∇²b − κρ` steady state, `Paper_GR-III`) from the opposite direction, with ED grounding the inputs of the thermodynamic one (severance = the hidden DOF, grain = the cutoff, severance count = η; `Paper_GravityAsEquationOfState`).

6. Two corollaries — and the one the log forbids me to overstate

Corollary 1 — the shear channel is exactly GR. The horizon shear viscosity of a gravitational equation of state is `η_{\rm shear} = 1/(16\pi G)` in GR; in Einstein-aether it is rescaled by the spin-2 coefficient $\propto$ `(1 − c_{13})`. On ED's luminal family `c_{13} = 0`, so the factor is unity: **ED adds no shear viscosity** (and the spin-1 dissipation is absent because hypersurface-orthogonality removes the vector mode). This is another face of "ED is observationally General Relativity": the luminal tensor cone forbids an aether shear footprint. *(Literature-grounded: Berglund–Bhattacharyya–Mattingly arXiv:1210.4940; luminal-family evaluation.)*

Corollary 2 — a scalar-sector co-suppression (stated at its honest tier, NOT a "fourth face"). The one genuinely new thermodynamic footprint is a **bulk (spin-0 / expansion) aether entropy-production term**, d_iS, absent from the equilibrium entropy (which is pure area) and appearing only in the dissipative/global sector. Its prefactor is the Noether current $J^a = -2c_{123}\,\vartheta\, f^a$ with $c_{123} = c_1 + c_2 + c_3 = c_{13} + c_2 = 0 + \lambda = \lambda$ on the luminal family — so it is $O(\lambda)$, the *same leading order* as the kinematic $\alpha_1 = -4c_{14} \approx -4\lambda$, and regular at $c_{13} = 0$ (no $1/c_{13}$). (Both are $\propto \lambda$ on the 1D family with $O(1)$ coefficients unresolved — pinning those is exactly the open computation below.) Both are the khronon (scalar) sector's expression, controlled by the one surviving luminal coupling.

The honest content, and its limit:

> **Banked:** on ED's derived luminal family (1D coupling space), the kinematic preferred-frame coupling α_1 and the bulk thermodynamic d_iS are **both $O(\lambda_{local})$**, each vanishing at least linearly in the GR limit ($\lambda \to 0$), so **neither leaks at $O(1)$** and both carry the same $\sim\rho_{event}/\rho_{Planck}$ sparse-becoming suppression (the thermodynamic footprint at least as suppressed as α_1's ~70 orders). The feared $O(1)/\Lambda$-sourced aether-entropy leak is refuted — it vanishes in the GR limit (Arata–Liberati–Neri arXiv:2603.28851). *(Well-supported; one inferential step — the linear-in-coupling flux prefactor, not a fully-rendered closed-form $S_æ(c_i)$.)*
>
> **NOT banked — the open conjecture:** whether α_1 and d_iS share the *exact same power* of λ (a genuine single-object "fourth face") or the entropy is *steeper* (e.g. λ^2, hence **more** suppressed and *not* unified) requires the closed-form khronometric aether entropy — which the 2026 covariant-phase-space literature did not render — plus a regularity check at the exact $c_{13} = 0$ point. Once the space is collapsed to 1D, "both are functions of λ" is automatic; the *unified-object* framing is rhetoric until the exact power is shown. The strong "fourth face / one structure" reading is therefore held as a conjecture, not a result.

This is the same discipline the working log enforced: the equilibrium result is the headline; the co-suppression is a real but tiered corollary; the fourth face is a flagged open question.

7. Honest tiers and the one open computation

- **Derived (in this article):** the local-Rindler Clausius steps — Raychaudhuri focusing, the boost-flux heat, the $\delta Q = TdS \to G_{ab} = 8\pi G_æ T_{ab}^{\rm tot}$ algebra (§2, §5).
- **Literature-grounded (cited, assembled):** the luminal-cone kinematic survival (§3); the pure-area Wald entropy and local reabsorption of the aether entropy (§4, Determination C); the shear-channel result (§6, Corollary 1).
- **Inference-tier:** the $O(\lambda_{local})$ power of the bulk d_iS and the global aether entropy (§6) — resting on the linear-in-coupling flux prefactor, not a rendered closed form.
- **Inherited:** the absolute magnitude of $\lambda_{local} \sim \rho_{event}/\rho_{Planck}$ (GR-IV estimate-tier); the grain value; the horizon 2π (via the Euclidean route, BH_Thermal2Pi).
- **Open (the one paper-able computation this leaves):** render the closed-form khronometric aether entropy on the luminal family, extract its **exact leading power in λ**, and check $c_{13} = 0$ regularity. This decides whether the scalar-sector co-suppression is "same power" (the fourth face is real) or "entropy even more suppressed" (still seventy-orders safe, but not a unified object). Until then, §6's conjecture stays a conjecture.

8. Position statement

Jacobson showed the Einstein equation is an equation of state. The obvious worry for Event Density — that its preferred foliation, the khronon, would obstruct a derivation built on local Lorentz invariance — turns out to be benign, and this article writes out why, step by step, for the first time in a khronometric theory. On ED's *derived* luminal family the causal scaffolding is the standard light cone (no universal horizon, Unruh survives, $\alpha_2 = 0$ exact), the horizon entropy is pure area (the aether kinetic term carries no curvature, and its would-be non-area piece reabsorbs in the local single-scale wedge), and so Jacobson's clean **equilibrium** Clausius relation runs unchanged and yields the **full Einstein equation** — the same equilibrium class as General Relativity, sharing GR's universal shear-viscosity term and adding none. The khronon leaves exactly two marks, and both are the scalar sector's: the kinematic α_1 and a bulk entropy-production `d_iS`, each `O(λ_local) ~ ρ_event/ρ_Planck` (the dissipative one at least as suppressed as α_1), seventy orders silent. Whether those two marks are the *same* function of the one coupling — a genuine "fourth face" of sparse becoming — is the one honest open computation this derivation leaves on the table, and it is named, not claimed. None of this is offered as confirmed physics; it is the grounding the equation-of-state program was waiting for: the arrow is in the law, absent from the equilibrium equation of state, and present only in a footprint the same sparseness that makes ED quantum keeps seventy orders quiet.

Cross-references

- Jacobson, T., *Phys. Rev. Lett.* **75**, 1260 (1995), gr-qc/9504004.
- **The verdict this writes out:** `event-density/foundations/Khronon_vs_Lorentz_in_Jacobson_RESULTS_LOG_` `07-24.md` (R1–R6, the closed analysis); `physics-papers/substrate-evaluation/Paper_GravityAsEquationOf` §6–§7 (the companion that asserts it).
- **ED gravity line:** `Paper_GR-II` (khronometric class, $b \to 0$ horizon), `Paper_GR-III_DynamicalRule.md` (`c_s = c`; §7.4 measured pure-area severance law), `Paper_GR-IV_ArrowsAlibi.md` ($\alpha_1 = -4\lambda_{local}$, $\alpha_2 = 0$, luminal 1D coupling family).
- **Horizon thermodynamics:** `foundations/BH_Thermal2Pi_FromNearHorizonRindler.md` ($T = \kappa/2\pi$); `foundations/BH_EntropyCoefficient_FromEventCounting.md` (the coefficient).
- **External (equation-of-state / aether thermodynamics):** Eling–Guedens–Jacobson gr-qc/0602001 (nonequilibrium spacetime); Chirco–Liberati PRD 81, 024016 (horizon shear viscosity); Brustein–Gorbonos–Hadad arXiv:0712.3206 (Wald entropy of higher-derivative/aether); Berglund–Bhattacharyya–Mattingly arXiv:1210.4940, arXiv:1309.0907 (universal-horizon thermodynamics); Foster gr-qc/0510125 (aether energy/first law); Arata–Liberati–Neri arXiv:2603.28851, arXiv:2606.27437 (covariant phase space; reabsorbability of the aether flux term); Barausse–Jacobson–Sotiriou arXiv:1104.2889, Blas–Sibiryakov arXiv:1110.2195 (universal horizons require superluminal modes); khronometric PPN: Blas–Lim arXiv:1412.4828, Ramos–Barausse arXiv:1811.07786, Foster–Jacobson gr-qc/0509083.
- **Companion (this family):** `Paper_GravityAsHorizonEquipartition.md` (Padmanabhan), `README.md`.